

## 5ARF0 Study Guide 2026

### Ethics of AI: Theory and Practice

#### Class Schedule (See below for readings details)

| Date/ time/ type          | Topic, Activity, or Deadline                                                                              | Readings (see list)                               |
|---------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| 1-Sep/ H5-6/<br>Lecture   | Introduction to ethics, human rights and AI                                                               | [UDHR]                                            |
| 1-Sep/ H7-8/<br>Tutorial  | Group formation and focus on multi-disciplinarity                                                         |                                                   |
| 3-Sep/ H1-2/<br>Lecture   | Philosophy of AI                                                                                          | [CHLEP Ch.1, Sections 1.1-1.3 & 1.10-1.13; Ch. 2] |
| 7-Sep                     | Online Q&A for topics ANWE (ONLINE 15:00-16:00), EADS (ONLINE 17:30-18:15), and ENSU (ONLINE 18:30-19:15) |                                                   |
| 8-Sep/ H5-6/<br>Lecture   | Ethics of AI                                                                                              | [CHLEP Ch. 3]                                     |
| 8-Sep/ H7-8/<br>Tutorial  | Literature search/ Assignment orientation                                                                 |                                                   |
| 10-Sep/ H1-2/<br>Lecture  | Fairness and AI                                                                                           | [CHLEP Ch. 4]                                     |
| 10-Sep/ H3-4              | First Round of Student Meetings                                                                           |                                                   |
| 14-Sep/ H9-10             | First Round of Student Meetings                                                                           |                                                   |
| 15-Sep/ H5-6/<br>Lecture  | Responsibility and Responsible AI                                                                         | [CHLEP Ch. 5]                                     |
| 15-Sep/ H7-8/<br>Tutorial | First Round of Student Meetings                                                                           |                                                   |
| 17-Sep/ H1-2/<br>Lecture  | Sustainability and AI                                                                                     | [CHLEP Ch. 6]                                     |
| 17-Sep/ H3-4              | First Round of Student Meetings                                                                           |                                                   |
| 18-Sep 23:59              | Assignment Phase 1 Due                                                                                    |                                                   |
| 22-Sep/ H5-6/<br>Lecture  | <b>Guest Lecture</b> Carlos Zednik: Explainable AI and Transparency                                       | Zednik 2021                                       |
| 22-Sep/ H7-8/<br>Tutorial | Second Round of Student Meetings                                                                          |                                                   |
| 24-Sep/ H1-2/<br>Lecture  | Human-Centered AI and Meaningful Human Control                                                            | Santoni de Sio & van den Hoven 2018               |
| 24-Sep/ H3-4              | Second Round of Student Meetings                                                                          |                                                   |
| 28-Sep/ H9-10             | Second Round of Student Meetings                                                                          |                                                   |
| 29-Sep/ H5-6/<br>Lecture  | Second Round of Student Meetings                                                                          |                                                   |
| 29-Sep/ H7-8/<br>Tutorial | Second Round of Student Meetings                                                                          |                                                   |
| 1-Oct/ H1-2/<br>Lecture   | Machine Ethics and Autonomous Systems                                                                     | Goodall 2014; Awad 2020                           |
| 6-Oct/ H5-6/<br>Lecture   | Superintelligence (Nickel)                                                                                | Chalmers 2010 Sections 1, 2, 5, 6, 7              |

|                           |                                                       |  |
|---------------------------|-------------------------------------------------------|--|
| 6-Oct/ H7-8/<br>Tutorial  | Work time (Nickel)                                    |  |
| 8-Oct/ H1-2/<br>Lecture   | No session                                            |  |
| 8-Oct/ H3-4               | Third Round of Student Meetings with Practice Pitches |  |
| 11 Oct 23:59              | Submit Draft of Phase 2 for Peer Review               |  |
| 12-Oct/ H9-10             | Third Round of Student Meetings with Practice Pitches |  |
| 13-Oct/ H5-6/<br>Lecture  | Third Round of Student Meetings with Practice Pitches |  |
| 13-Oct/ H7-8/<br>Tutorial | Third Round of Student Meetings with Practice Pitches |  |
| 15-Oct/ H3-4/<br>Lecture  | Third Round of Student Meetings with Practice Pitches |  |
| 16 Oct 23:59              | Complete Peer Review (Individual)                     |  |
| 20-Oct/ H5-6              | Office hour                                           |  |
| 20-Oct 23:59              | Submit Video Pitch                                    |  |
| 22-Oct/ H1-2              | Exam review                                           |  |
| 23 Oct 23:59              | Submit Phase 2 Final Version                          |  |
| 24 Oct 23:59              | Complete Groupmate Evaluation                         |  |

**[UDHR]** United Nations, Universal Declaration of Human Rights. (1948). Available at <https://www.un.org/en/about-us/universal-declaration-of-human-rights>

**[CHLEP]** Smuha NA, ed. (2025). *The Cambridge Handbook of the Law, Ethics and Policy of Artificial Intelligence*. Cambridge Law Handbooks. Cambridge University Press.  
<https://www.cambridge.org/core/books/cambridge-handbook-of-the-law-ethics-and-policy-of-artificial-intelligence/contents/B58479AF2A9C398BD461575A18F828D3>

Awad, E., Dsouza, S., Bonnefon, J. F., Shariff, A., & Rahwan, I. (2020). Crowdsourcing moral machines. *Communications of the ACM*, 63(3), 48-55.

Chalmers, D.J. (2010). The Singularity: A Philosophical Analysis. *Journal of Consciousness Studies* 17: 7-65. Available at <https://consc.net/papers/singularityjcs.pdf>

Goodall, N. J. (2014). Ethical Decision Making During Automated Vehicle Crashes. *Transportation Research Record: Journal of the Transportation Research Board*, No. 2424, Transportation Research Board of the National Academies, Washington, D.C., 2014, pp. 58–65. DOI: 10.3141/2424-0

Santoni de Sio, F., & Van den Hoven, J. (2018). Meaningful human control over autonomous systems: A philosophical account. *Frontiers in Robotics and AI*, 5, 15.

Zednik, C. (2021). Solving the Black Box Problem: A Normative Framework for Explainable AI. *Philosophy & Technology* 34: 265-288.

There are also recommended readings for each of the project topics, which can be found in the Case Description Packet.

## Official course information from OSIRIS:

### 1. Basic course information

|                         |                                          |
|-------------------------|------------------------------------------|
| Academic year           | 2026-2027                                |
| Quartile                | 1, timeslot E                            |
| Course code             | 5ARF0                                    |
| Course name             | Ethics of AI: Theory and Practice        |
| Credits (ECTS)          | 5                                        |
| Course category         | not applicable                           |
| Course type             | graduate school                          |
| Target group            | master AI&ES students                    |
| Language of instruction | English                                  |
| Research group          | IE&ES groups: Philosophy and Ethics 100% |
| Responsible lecturer    | dr. Philip Nickel                        |
| Contact person          | dr. Philip Nickel                        |
| Co-lecturer(s)          | TBA                                      |

## **2. Motivation**

Core course in master program AI&ES

## **3. Aims / Learning goals**

After following this course, the student is able to:

Recognize and assess ethical problems in AI and develop a plan to find solutions in cooperation with stakeholders.

Identify ethical issues and value conflicts at different stages of the design process.

Describe these issues and conflicts as they relate to intelligent systems, using ethical frameworks and concepts.

Adopt a strategy for resolving value conflicts in a way that is convincing to multidisciplinary stakeholders.

Use these reflections to justify choices in the design of intelligent systems and interfaces.

## **4. Contents**

### **Description of the course contents**

This course addresses the human and ethical impacts of AI, giving students concepts and approaches to understand how AI and humans interact and to anticipate ethical issues that arise due to the application of AI to real-life problems.

AI affects human behavior and vice versa: AI can support humans in repetitive, easy to automatize tasks, reducing human error and boredom, as well as for complex tasks, extending their cognitive abilities, and improving or augmenting human decision making. However, AI can lead to job erosion, loss of skills, overreliance on automation, and other effects that should be prevented or mitigated.

Students learn to give reasoned answers to central ethical questions about AI such as: What is an ideal vision for the future of AI and automation? How can we assure that people

affected by AI have meaningful control over artificial intelligence and robotics? How can professionals working in the field respond to ethical challenges that occur in their work? Topics include: ethical principles and theories, meaningful human control, moral responsibility, trust in AI, opacity and explainability, ethics of generative artificial intelligence.

**ECTS breakdown: description of the ECTS distribution over the different content elements and how much time a student will spend on each content element**

2.5 ECTS: students acquire theoretical understanding of ethics concepts and their application to AI.

Following lectures, reading and supervised tutorials, self-study, taking practice quizzes and exam

2.5 ECTS: students motivate an approach to a real-life problem case involving ethics and AI, and develop a strategy for addressing it with stakeholder input.

Supervised tutorials, student study group meetings, Internet research, writing

**5. Instruction methods (describe lab work separately)**

Lectures, supervised tutorials, self-study, practice quizzes, presentations  
Student study groups focusing on the written assignment

**6. Materials**

- a. **Required materials.** None
- b. **Recommended materials** See Literature above

**7. Position in the curriculum**

- a. **Entrance requirements.** None
- b. **Assumed previous knowledge.** Basic ethics concepts
- c. **Assumed previous knowledge can be gained by.** OSAB0 USE basic: ethics and history of technology or ITEC
- d. **Follow up courses.** 0HM340 Human-AI Interaction
- e. **Is the course part of a coherent package? If so, please state which coherent package.** No

**8. Assessment method (tests and weights)**

Written assignment (50%) & written (essay) exam (50%)

The written assignment is an interim assignment in which the students go into detail motivating an approach to a real-life problem, based on an analysis of scientific literature, experimentation with solutions, and consultation with domain experts.

There will be three possible topics, which will be further specified by each group:

**ANWE:** AI for Animal Welfare

**ENSU:** AI for Environmental Sustainability

**EADS:** AI for Eating Disorder Symptoms

The written assignment will be assessed primarily on awareness of ethical concerns raised by different approaches, motivation for iterative solutions to the problem, and sophistication in identification and anticipation of limitations and barriers in the target context. Clarity and compellingness of the design solution will also be assessed.

The written assignment includes Phase 1 which is a first version (10%). It also includes a peer review between groups, a presentation (10%) and an intra-group peer evaluation and reflection (10%).

Written (essay) exam means an on-campus exam, in which the students are expected to write out their answers (i.e., most answers will be open format). The exam tests competency in identifying and applying ethics and psychology concepts from the lectures. An example would be the ability to explain, apply, and critique a notion of trust, fairness, or manipulation as applied to AI systems. Students might be asked to reproduce central concepts from the lectures, or they might be given a case to which they must apply one or more concepts.

Students must obtain a grade of at least 5,0 on the written exam in order to pass the course.

## 9. Other information. See below.

### Academic Integrity

#### Ground rules for students

Be aware that you should always operate with academic integrity, taking into account the rules and regulations about copyright, plagiarism and fraud. Use the links below to learn more about what you are (not) allowed to do in your assignments and assessments.

|                                                      |                                                   |
|------------------------------------------------------|---------------------------------------------------|
| <a href="#">TU/e Rules for AI use in Education</a>   | <a href="#">EE BSc Regulations and PER</a>        |
| <a href="#">TU/e Regulations and Code of Conduct</a> | <a href="#">EE MSc Regulations and PER</a>        |
| <a href="#">TU/e Copyright Information Point</a>     | <a href="#">AI&amp;ES MSc Regulations and PER</a> |

#### On the use of AI

Any work for an assignment in which generative AI tools are allowed, should make unambiguously clear:

- which tools you used,
- for which parts of the work you used them,
- what you used them for.

You should be critical about the quality of your information sources and make sure to precisely reference the parts of your work where AI tools were used.

**In any use of AI, you remain responsible for the work you deliver, down to the individual sentence.**

**Use of AI other than that indicated may lead to a suspicion of fraud.**

AI is the object of your assignment, not the process by which you do the assignment. AI is not a wise means of explaining your process of working with that object or justifying your

thought process. You may use it in superficial ways to brainstorm ideas or do a spelling and editing check, at your own risk. You should maintain complete critical control over its use.

For this reason, you are not allowed to use generative AI to compose the text of your assignment. You are allowed to use AI tools as a source for inspiration or to improve/enhance the quality of products, although this is not recommended. If you do choose to use these tools, you should be able to explain why you opted to use specific tools and evaluate their usefulness.